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SUPERSEDING
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FEDERAL SPECIFICATION

FITTINGS, PLASTIC PIPE
(ADAPTERS, COUPLINGS, ELBOWS AND TEES FOR POLYETHYLENE PIPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers plastic insert fittings for polyethylene (PE) plastic pipe and a straight metal adapter for connecting PE pipe to steel pipe.

1.2 Classification.

1.2.1 Types and styles. Fittings shall be of the following types and styles, as specified (see 6.2):

Types I - Adapters.

Style A - NPS male end.
B - NPS female end.

Type II - Couplings.
III - Elbows.
IV - Tees.

1.2.2 Size. Fittings shall be of the following nominal pipe sizes, as specified (see 6.2):

1/2, 3/4, 1, 1-1/4, 1-1/2, 2, 2-1/2, 3 and 4 inches.
1.3, 1.9, 2.5, 3.2, 3.8, 5.1, 6.4, 7.6 and 10 cm.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

WW-C-440 - Clamps, Hose, (Low-Pressure).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. .

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 4730

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of the specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standard for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

American Society for Testing and Materials (ASTM) Standards:

D1599 - Test for Short-Time Rupture Strength of Plastic Pipe, Tubing, and Fittings.
D2609 - Specification for Plastic Insert Fittings for Polyethylene (PE) Plastic Pipe.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Sanitation Foundation Testing Laboratory, Inc., (NSF) Standard:

No. 14 - Thermoplastic Materials, Pipe, Fittings, Valves, Joining Materials.

(Application for copies should be addressed to the National Sanitation Foundation Testing Laboratory, Inc., P.O Box 1468, Ann Arbor, MI 48106.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Plastic fittings. Unless otherwise specified (see 6.2), fittings shall be made from any of plastic materials listed in ASTM D2609.

3.1.2 Metal adapters. Metal adapters shall be steel. Composition of the steel shall be at the manufacturers option.

3.1.3 Clamps. Clamps shall be corrosion-resisting steel. Unless otherwise specified (see 6.2), corrosion-resisting steel shall be AISI 200, 300, or 400 series at the manufacturer's option.

3.2 Plastic elbows, tees and adapters. These fittings, with insert ends and without clamps shall be in accordance with ASTM D2609 (see 4.3.1 and 4.3.2).

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3.3 Plastic couplings. Couplings shall be with insert ends, and without clamps. Insert ends shall have annular notches for tight grip inside pipe walls. Dimensions of insert ends shall be in accordance with ASTM D2609. Couplings shall have a raised outside annular stop midway between insert ends. Dimensions of the stop and length of coupling shall be at the manufacturer's option.

3.4 Public health and safety hazards, plastic fittings. When specified (see 6.2), plastic fittings intended for transporting potable water shall meet the following requirements:

3.4.1 Each contractor shall submit to the contracting agency proof that the fittings he proposes to supply under this specification conform to the requirements of the National Sanitation Foundation Standard No. 14. The label, or listing with reexamination, of the National Sanitation Foundation may be accepted as evidence that PE fittings conform to this requirement.

3.4.2 In lieu of the label, or listing with reexamination, the contractor may submit independent proof, satisfactory to the contracting agency, that the fittings conform to the applicable requirements of the published standards, including methods of test, of the National Sanitation Foundation Standard No. 14.

3.4.3 Compliance with the above preliminary requirements in regard to public health and safety hazards does not absolve the bidder from complete compliance with the requirements of the specification in order to secure acceptance of his material.

3.5 Color plastic fitting. Color of plastic fittings shall be white, black, or any other color, as specified (see 6.2). Unless otherwise specified (see 6.2), value of white and black, and value, hue and chroma of other colors shall be at the manufacturer's option.

3.6 Metal adapters.

3.6.1 Construction. Steel adapters shall be seamless or seam welded. Dimensions of insert end for PE pipe shall be in accordance with ASTM D2609. Threaded end for coupling to steel pipe shall have external threads, and threads shall conform to pipe threads in NBS H28.

3.6.2 Finish. Steel adapters shall be zinc coated or cadmium plated, as specified (see 6.2). Zinc coat shall be electroplated or hot dipped (galvanized) at the manufacturer's option. Unless otherwise specified (see 6.2), thickness of cadmium electroplate, and thickness or weight of zinc coat shall be at the manufacturer's option.

3.7 Clamps. Clamps shall be type F, worm screw tangentally attached with slotted machine screw, in accordance with WW-C-440. Clamp size shall be as follows and as specified (see 6.2):

Pipe size		Clamp size	
Inches	cm	Inches	cm
1/2	1.3	9/16	1.4
3/4	1.9	29/32	2.3
1	2.5	1-1/4	3.2
1-1/4	3.2	1-1/2	3.8
1-1/2	3.8	2	5.1
2	5.1	2-3/4	7.0
2-1/2	6.4	2-5/16	5.9
3	7.6	3-1/2	8.9
4	10	4-1/2	11

3.8 Workmanship. Plastic fittings and steel adapters shall be free of cracks, holes, and inclusions. Plastic fittings shall be uniform in color. Coated steel adapters shall be free from uncoated areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection. Inspection levels and acceptable quality levels (AQL's) shall be in accordance with MIL-STD-105.

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4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of each type, style, and size of fitting submitted for inspection at the same time.

4.2.2 Dimensional and visual examination of end item. Fittings shall be examined for these characteristics in accordance with level S-3 and an AQL of 2.5 percent defective for major defects and 4.0 percent defective for minor defects listed in table I.

TABLE I. Classification of defects, end item

Defects	Major	Minor
Type and style not as specified.	X	
Material for plastic fittings not as specified when required.	X	
Cracks, holes, or inclusions in surface of fittings.	X	
Color of plastic fittings not as specified when required.	X	
Color of plastic fittings not uniform.		X
Uncoated areas on metal adapters.	X	
Size of clamps not as specified when required.	X	
Dimensions not within tolerance.	X	
Alignment of all openings of fittings not within 1/4 inch per foot (6.4 mm per 0.31 m).	X	
Threads and barbs not uniform.	X	
Marking missing or illegible.		X
Damage or defects affecting function or serviceability.	X	
Damage or defects not affecting function or serviceability.		X

4.2.3 Samples for test. Sampling of fittings for tests shall be in accordance with level S-2. The AQL for tests shall be 4.0 percent defective.

4.2.4 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing and marking requirements comply with section 5.

4.3 Test method.

4.3.1 Specimens, plastic fittings. Prepare fittings for test in accordance with ASTM D2609 (see 3.2).

4.3.2 Pressure test, plastic fittings. Test for burst pressure in accordance with ASTM D1599 (see 3.2).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Levels A and B. Ten fittings shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665 or PPP-B-676. Each box shall be secured to prevent accidental opening.

5.1.2 Level C. Fittings shall be packaged in accordance with the supplier standard practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Unit packages, in quantities as specified (see 6.2), shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed, sealed and reinforced in accordance with the appendix to PPP-B-636.

5.2.2 Level B. Twenty-four unit packages (240 fittings) of one stock number, shall be packed in a box conforming to PPP-B-636. The box shall be closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C. Fittings shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with National Motor Freight Classification or Uniform Freight Classification rules.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military agencies. In addition to any special markings required by the contract or order, all interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Fittings are intended for use with PE pipe.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specifiation.
(b) Type, style and size (see 1.2.1 and 1.2.2).
(c) Material for plastic fittings, if other than specified (see 3.1.1).
(d) Clamps, type of corrosion-resisting steel, when required (see 3.1.3).
(e) Whether plastic fittings are to be used for potable water (see 3.4).
(f) Color of plastic fittings, when required (see 3.5).
(g) Finish on metal adapter, and when required coating thickness for cadmium electroplate, and thickness or weight of zinc coat (see 3.6.2).
(h) Clamps, size when required (see 3.7).
(i) Size of lot, if different from 4.2.1.
(j) Selection of applicable levels of packaging and packing required (see 5.1 and 5.2).
(k) Marking (see 5.3).

6.3 Metric equivalents. The metric equivalents to dimensional requirements are for information only and are not intended to be used for determining acceptance or rejection of items otherwise conforming to the requirements of this specification.

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity:

GSA-FSS

Civil Agency Coordinating Acticities:

COMMERCE - NBS
HEW - FEC
INTERIOR - BPA

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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